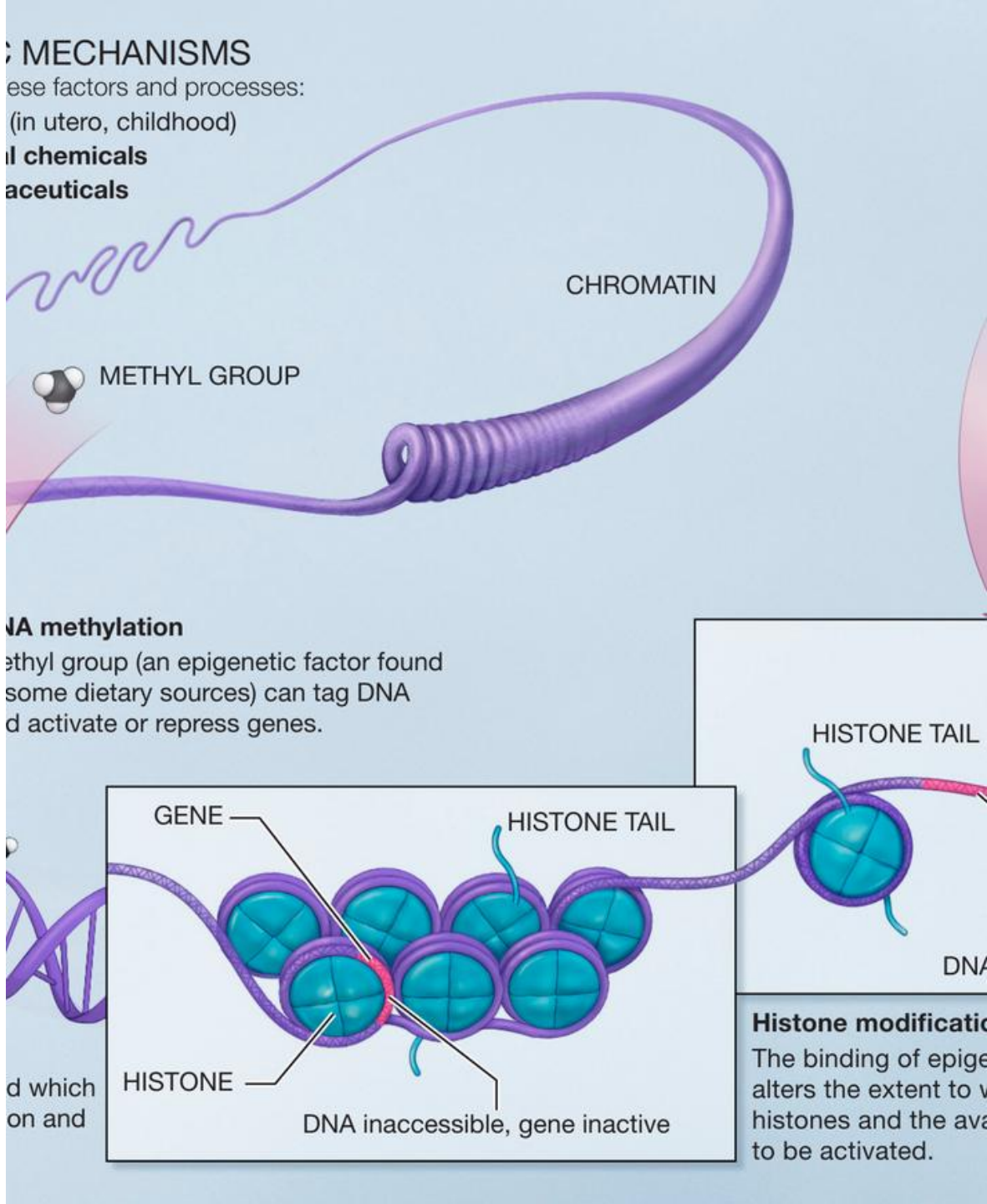
DP Biology – D2.2 (HL only)

Gene expression and epigenetics

- How can a cell change which genes it expresses?
- How are patterns of gene expression passed on?

HL statements D2.2.1–11. Textbook pp. 670–680.
Image: epigenetic mechanisms. National Institutes of Health, public domain.

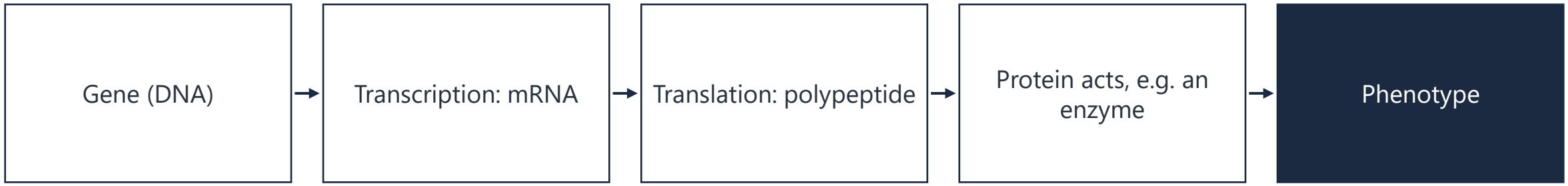


Objectives for each class

Class	Statements	By the end of the class, you can...
1	D2.2.1–5	Explain how gene expression is controlled by transcription factors and mRNA breakdown, and compare genome, transcriptome and proteome
2	D2.2.6–8	Explain methylation as an epigenetic tag, epigenetic inheritance, and effects of the environment
3	D2.2.9–11	Explain why tags are reset in gametes, evaluate twin studies, and explain how lactose and hormones switch genes

From gene to phenotype

D2.2.1



Example

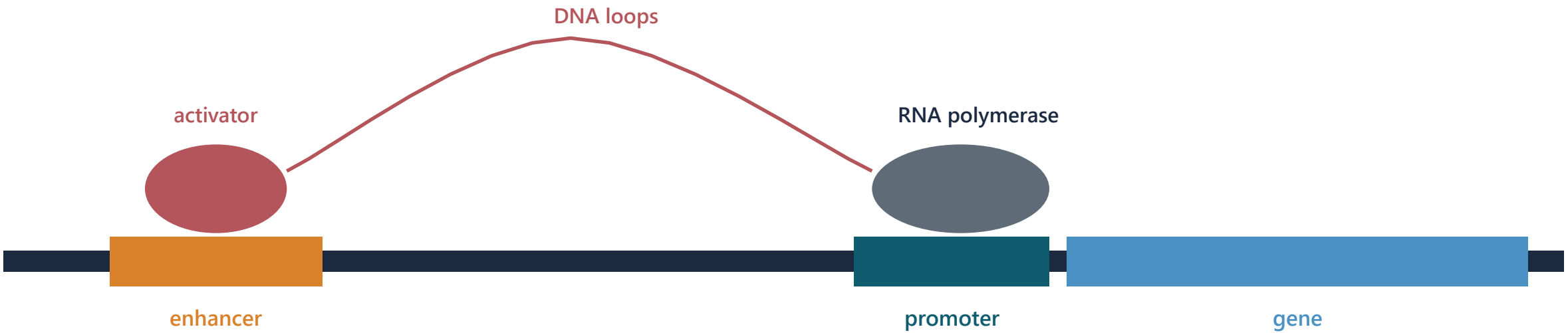
The tyrosinase gene is expressed in skin cells; the enzyme starts the pathway to melanin. A non-working allele gives albinism.

Key idea

Gene expression is how a gene's information affects the phenotype. Most genes are expressed only in some cells, at some times.

Controlling transcription

D2.2.2

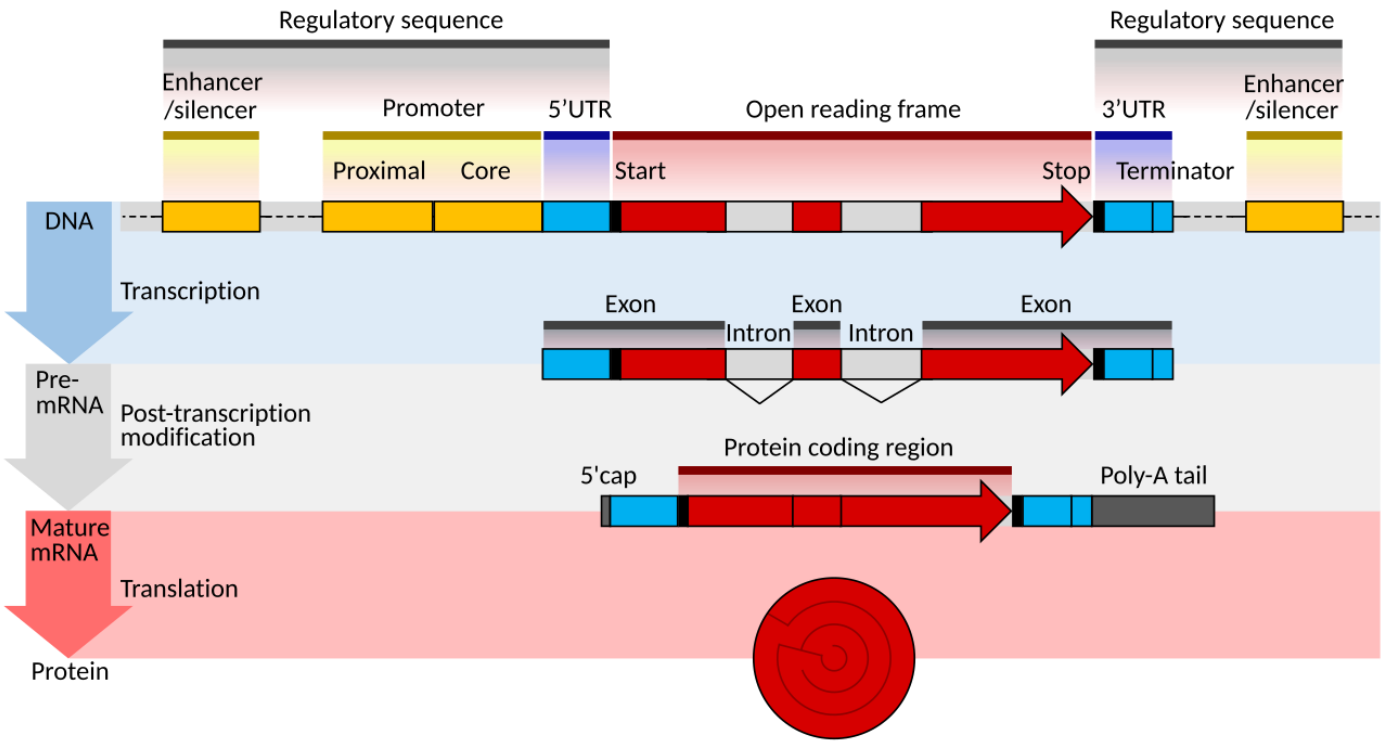


Transcription factors are proteins that bind specific DNA sequences. Some bind the **promoter**, helping RNA polymerase start. **Activators** bind **enhancers**, sometimes far away; the DNA loops so they reach the promoter. **Repressors** bind **silencers**.

Different cells hold different transcription factors, so they transcribe different genes.

Regulatory sequences of a gene

D2.2.2



A eukaryotic gene with its regulatory sequences. Thomas Shafee, CC BY 4.0.

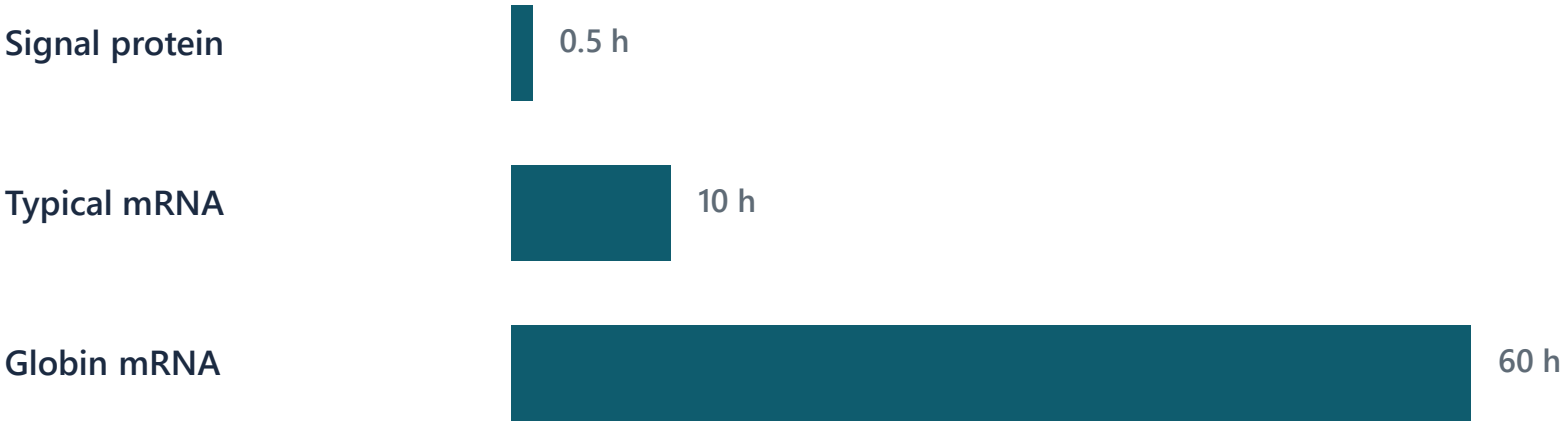
Sequence	Binds
Promoter	RNA polymerase
Enhancer	Activators
Silencer	Repressors

Think

Why might one gene have several enhancers, each used in a different tissue?

Controlling mRNA breakdown

D2.2.3



Typical mRNA lifetimes in human cells (rounded).

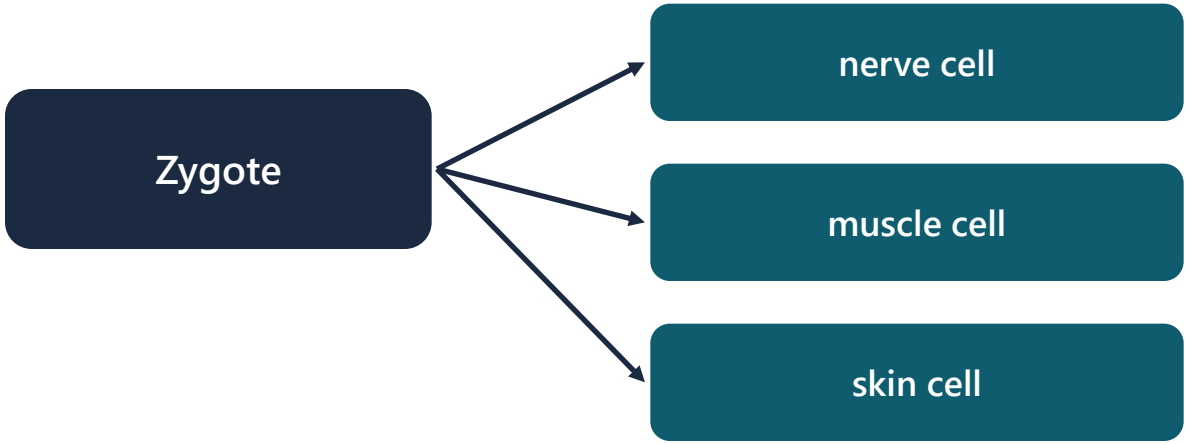
mRNA has a **poly-A tail** that is slowly shortened in the cytoplasm; then nucleases break it down. The longer mRNA lasts, the **more protein** is made.

Why break it down?

So a cell can stop making a protein. Short-lived mRNAs suit brief signals.

Epigenesis: building a body

D2.2.4



Epigenesis: as an organism develops, its cells become different by switching genes **on and off**.

The base sequence (genotype) is the same in every cell; gene expression, and so the cell's phenotype, differs.

Common mistake

Epigenetic changes are not mutations. A mutation changes the sequence; an epigenetic change alters which genes are read.

Genome, transcriptome, proteome

D2.2.5

	What it is	Varies?
Genome	All the genetic information	Same in every cell of an individual
Transcriptome	All the mRNA in a cell at one time	Differs between cells and over time
Proteome	All the proteins in a cell at one time	Differs; depends on the transcriptome

Key idea

No cell expresses all its genes. The pattern of expressed genes decides how a cell differentiates.

Practice: same genes, different cells

D2.2.1–5

Liver and nerve

Do they have the same genome? Transcriptome?

Remove an enhancer

Predict the effect on a gene in one tissue.

Long-lived mRNA

Predict the effect on the protein made.

Explain

Give a reason for each answer using the words transcription factor, transcriptome or translation.

Check-in: class 1

D2.2.1–5

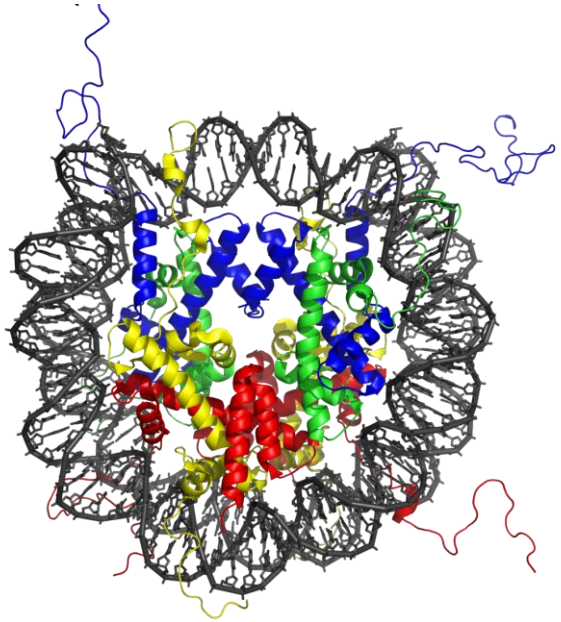
1. Define gene expression.
2. Explain how activators and enhancers raise transcription.
3. Explain how mRNA breakdown controls protein production.
4. Define epigenesis.
5. Distinguish the genome from the transcriptome.

How to use this

Answer without your notes, then check against slides 3–9.

Epigenetic tags: methylation

D2.2.6



Nucleosome. Zephyris, CC BY-SA 3.0.

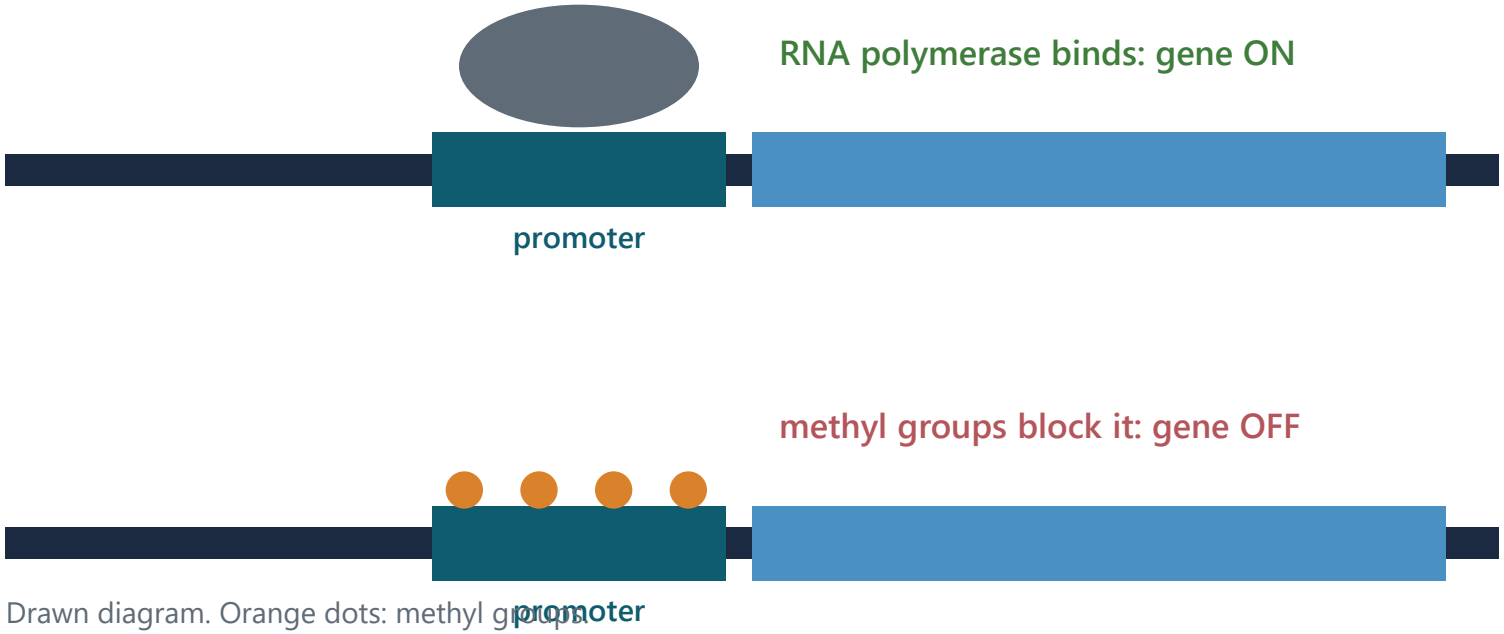
	DNA methylation	Histone methylation
Where	Cytosine in the promoter	Histone tails
Effect	Represses the gene	Can repress or activate

Other tags

Acetylation of histones loosens packing and usually activates genes. Enzymes add and remove tags, so they can change during life.

How a methyl tag silences a gene

D2.2.6



Drawn diagram. Orange dots: methyl groups.

Methyl groups on **cytosines** in a promoter stop transcription factors binding, and the DNA packs more tightly.

The sequence is unchanged.

Passing on gene expression

In mitosis, methylation patterns are **copied** to the new DNA, so a liver cell gives rise to liver cells.

If tags survive meiosis and fertilization, a change in a parent's gene expression can pass to offspring: **epigenetic inheritance**, with no change in sequence.

Dutch Hunger Winter

People whose mothers were pregnant during the 1944–45 famine showed methylation differences decades later, and more of some diseases. How strong is this evidence?

Think

Why is evidence from a natural event like a famine hard to interpret?

The environment changes gene expression

D2.2.8



Agouti mice. R. Jirtle and D. Dolinoy, CC BY 3.0.

Diet: in agouti mice, a mother's diet rich in methyl donors increases methylation of the agouti gene: brown, lean offspring instead of yellow, obese ones.

Air pollution is linked to methylation changes in immune genes, and to asthma.

Evaluate

Most studies show correlation, not cause. What evidence would show a pollutant causes a methylation change, and the change causes disease?

Working with data: diet and coat colour

D2.2.8

Offspring of genetically identical agouti mothers. The values are invented for practice; the pattern matches published studies.

Mother's diet	Yellow, obese / %	Brown, lean / %
Normal	70	30
Rich in methyl donors	35	65

1. Describe the effect of the diet.
2. Explain it using methylation.
3. Explain why the mothers must be genetically identical.

Check-in: class 2

D2.2.6–8

1. Explain how DNA methylation affects transcription.
2. State the effect of histone acetylation.
3. Explain how methylation patterns pass to daughter cells.
4. Define epigenetic inheritance.
5. Give two environmental factors that alter gene expression.

How to use this

Answer without your notes, then check against slides 11–15.

Resetting tags in eggs and sperm

Most tags are **removed** in cells that become gametes, and again in the early embryo. The zygote can then develop into every cell type.

Some tags survive: in **imprinted** genes the copy from one parent is silenced, so expression depends on which parent it came from.



A liger. Dinkun Chen, CC BY-SA 4.0.

Ligers and tigons

Liger: lion father, tiger mother: huge. Tigon: tiger father, lion mother: similar or smaller.

Why ligers are so big

D2.2.9

	Lion	Tiger
Mating	Females mate with several males	Usually one male
Father’s imprint	Growth genes active	Little growth promotion
Mother’s imprint	Growth-limiting genes active	Little limitation needed

Put together

A liger gets growth-promoting tags from its lion father but no balancing limits from its tiger mother, so it grows very large. This hypothesis fits the evidence; the genes are not yet known.

Monozygotic twin studies

D2.2.10

Identical twins come from one zygote: the **same genome**. Differences between them come from the environment, including epigenetic changes.

Young twins have very similar methylation; older twins differ more, especially if their lives differ.

Evaluate

Twins share a womb and often a home. Twins raised apart help separate the effects. Small samples and self-reported lifestyles are limits.

Working with data: methylation in twins

D2.2.10

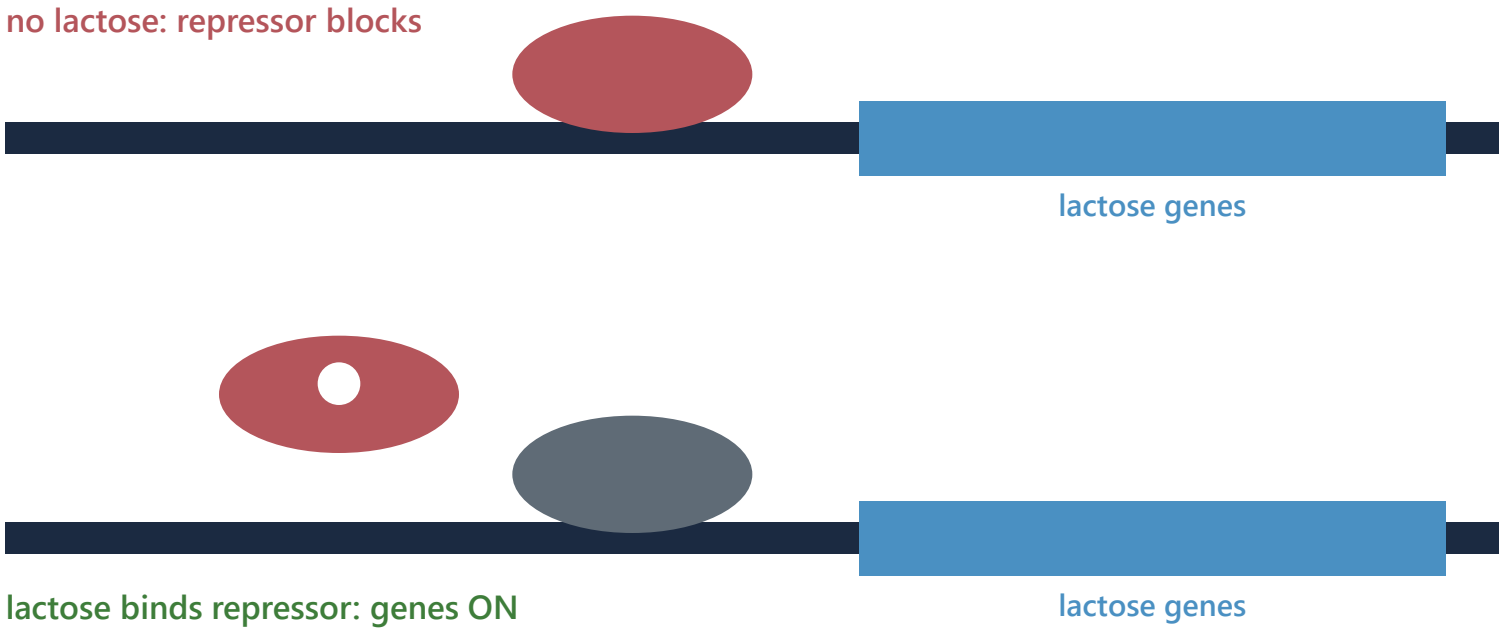
Difference in DNA methylation between identical twins (relative units). Invented for practice; the pattern matches published studies.

Age / years	Difference
3	1.0
10	1.6
28	2.9
50	4.8

1. Describe the trend.
2. Suggest two reasons twins diverge with age.
3. Explain why twins help separate genes from environment.

Lactose switches genes in E. coli

D2.2.11



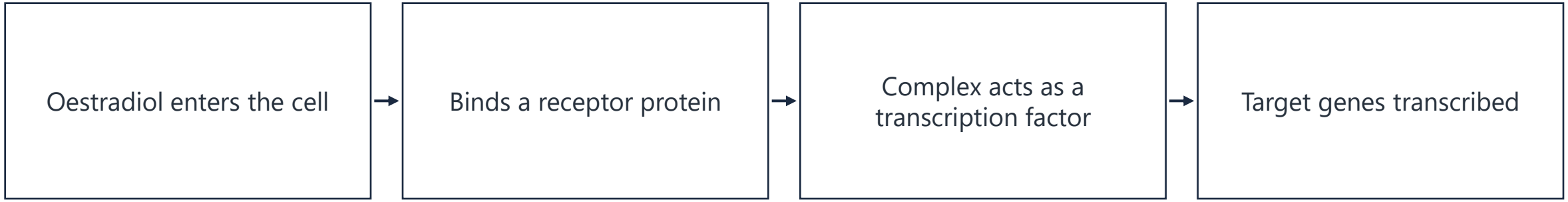
With no lactose, a **repressor** sits on the DNA. Lactose binds the repressor so it falls off, and the genes for taking up and digesting lactose are transcribed.

No wasted enzymes when there is no lactose.

Drawn diagram. White dot: lactose. Grey: RNA polymerase.

A hormone switches genes

D2.2.11



Steroid hormones pass through the membrane. Oestradiol switches on genes such as the **progesterone receptor** gene in target cells.

One signal changes gene expression in many cells at once.

Compare

Lactose acts on a bacterium's own genes; a hormone carries a signal between organs. Both change the pattern of gene expression.

Practice: explain the observation

D2.2.9–11

No lactose

E. coli on glucose makes almost no lactose enzymes.

Older twins

Twins at 60 differ more than twins at 5.

Tigon

A tigon is no bigger than its parents.

For each one

Explain using gene expression, epigenetic tags or imprinting.

Check-in: class 3

D2.2.9–11

1. Explain why most tags are removed in gametes.
2. Define genomic imprinting.
3. Explain why a liger grows larger than a tigon.
4. Explain what twin studies show about gene expression.
5. Explain how lactose switches on genes in E. coli.

How to use this

Answer without your notes, then check against slides 17–23.

Common mistakes to avoid

Not this	But this
Epigenetic changes alter the base sequence.	They change expression, not sequence.
Every cell expresses every gene.	Each cell type expresses a different set.
Methylation always switches genes on.	Promoter methylation usually switches genes off.
Identical twins are identical in every way.	Same genome; epigenetic patterns diverge.

Summary: gene expression

	Key idea
D2.2.1–3	Transcription factors and mRNA lifetime control expression
D2.2.4–5	Epigenesis; genome fixed, transcriptome and proteome vary
D2.2.6	DNA and histone methylation act as tags
D2.2.7–8	Tags copied in mitosis; diet and pollution change them
D2.2.9	Tags reset in gametes; imprinting; ligers
D2.2.10	Twins: same genome, diverging methylation
D2.2.11	Lactose and oestradiol switch genes

Check that you can...

1. **Explain** how gene expression links genotype and phenotype
2. **Explain** how transcription and mRNA breakdown are regulated
3. **Compare** genome, transcriptome and proteome
4. **Explain** methylation as an epigenetic tag
5. **Outline** epigenetic inheritance and environmental effects
6. **Evaluate** evidence from twin studies

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between the genome and the proteome of a cell.
2. Explain how methylation of a promoter affects a gene.
3. Suggest why a liger grows larger than a tigon.

Image credits

Epigenetic mechanisms.png – National Institutes of Health – Public domain
Gene structure eukaryote 2 annotated.svg – Thomas Shafee – CC BY 4.0
Nucleosome 1KX5 colour coded.png – Zephyris at English Wikipedia – CC BY-SA 3.0
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Liger – Shenzhen Safari Park 2023-02-03 01.jpg – Dinkun Chen – CC BY-SA 4.0

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